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A Recent Advancements in Multi-Modal Medical Image Fusion Techniques using SWT, NSST, NSCT, and CNN

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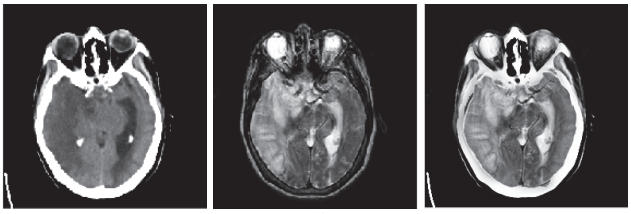
***Abstract*— Image fusion is a method used in image processing to provide a more complete representation by amalgamating features and data from many images. Multi-modal medical image fusion involves the integration of medical images from many imaging modalities, including computed tomography (CT) scans, positron emission tomography (PET), and magnetic resonance imaging (MRI), into one single dataset. This integration enhances the visualisation of anatomical structures and clinical situations, hence improving diagnostic accuracy by leveraging the strengths of each medium. This study employs MRI, CT, and PET scans as experimental modalities. This review aims to compare various multi modal medical image approach based on Stationary Wavelet Transform (SWT), Non-Subsampled Shearlet Transform (NSST), Convolutional Neural Network (CNN) and Non-Subsampled Contourlet Transform (NSCT). This study examines the latest conventional and non-conventional research conducted within these disciplines. It further evaluates these approaches according to diverse image quality metrics and many quantitative assessments. According to this comparison, CNN-based fusion demonstrates superior results, as the overall visual and parametric quality of the fusion outcomes surpasses that of the other approaches evaluated.**

Keywords—image fusion, multi-modal medical image fusion, PET image, CT image, MRI image.

I. INTRODUCTION

The majority of images just include one piece of information or very little. Examples include variations in focus locations and the blurring of objects that are either too near or too far away in one image. If various kinds of multi-modal images can be clearly recognised, then they are considered to be distinct kinds of images. Information is reflected in several kinds of images. It muddles the data to the point that the doctor can't make an informed decision. Many experts in medical diagnosis have taken an interest in this subject. The five essential phases required for effective image fusion are registration, decomposition, fusion rules, reconstruction, and assessment.

Fusion of images may be multi-focus, multi-sensor, multi-scale, multi-modal, multi-spectral, or multi-temporal; the specifics depend on the dataset. Sensor data is fused together in multi-sensor image fusion. The end result is an image that is both comprehensive and easy to understand. That is to say, it may go unnoticed by the average person. For precise abnormality localisation, data derived from combined images may be used. Various multi-modal medical image fusion algorithms are presented in this review study. In order to improve imaging quality while maintaining features, multi-modal medical image fusion combines several images from various imaging modalities. Current research issues in medical image fusion include computer vision, pattern recognition, machine learning, artificial intelligence, and image processing, among many others. Due to its ability to improve clinicians' understanding of the injured region by merging images from several medical modalities, medical image fusion has seen substantial clinical application. The example of multi-modal medical image fusion is shown in Fig 1.



(a) (b) (c)

Fig 1. Example of multi-modal medical image fusion (a) Source image 1: CT image (b) Source image 2: MRI image (c) Fused image

This research examines the integration of MRI, PET, and CT scan images to provide a more complete medical image. Each modality offers distinct knowledge unattainable via the others. An MRI image may provide insights into blood flow, tumors, vascular problems, and specifics about soft tissues, the brain and spinal cord, as well as joint and muscle ailments. Conversely, CT scans provide anatomical information on bones, lungs, and the thoracic, abdominal, and pelvic regions, facilitate injury diagnosis, and identify the presence of cancer, among other applications. Attributes include rapid scanning MRI images do not disclose slices, calcified tissues, or characteristics of bone. A significant drawback is radiation exposure, and the images acquired from CT scans do not provide the requisite diagnostic information to elucidate the characteristics of soft tissues, such as those found in the brain and muscles. Additionally, a CT scan cannot provide functional information on metabolic activity or blood flow, in contrast to MRI and PET scans. The functional parameters that may be identified using PET imaging include cellular activity, glucose metabolism, neurotransmitter activity, blood flow, and oxygen utilization. It may also be used for comprehensive body scans, facilitating early illness diagnosis, and tracking therapy progress. PET images are inadequate for distinguishing soft tissues, providing comprehensive

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anatomical information, and identifying structural changes in the body.

This review study aims to examine several multimodal medical imaging approaches using SWT, NSST, NSCT, and CNN, focusing on criteria such as objectives, methodologies, advantages, and disadvantages. This review article examines the latest conventional and non-conventional research conducted within these topics. It further evaluates these approaches according to diverse image quality metrics and many quantitative assessments.

After the introduction, Section 2 briefly describes methods based on various factors. Section 3 discuss the comparative results, and the review is concluded in Section 4.

II. Literature Survey

A. SWT Domain

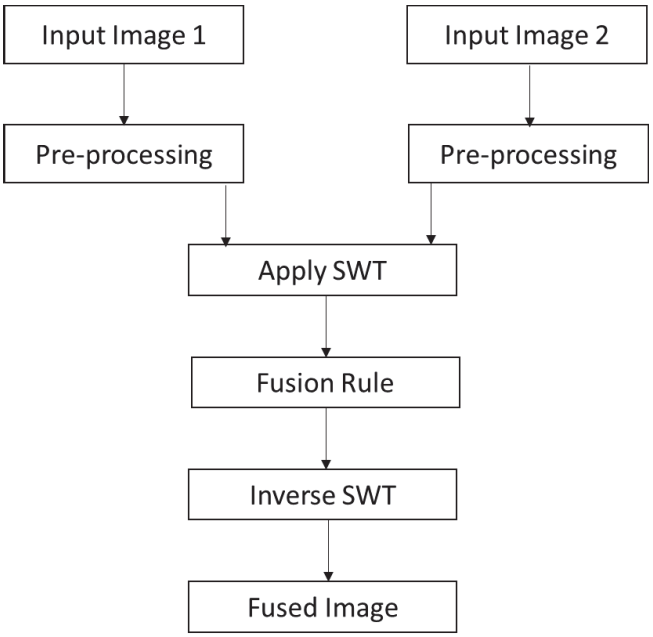


Fig 2. General image fusion process using SWT

The general image fusion process using SWT is shown in Fig 2. According to the Fig 2, input images 1, and 2 are pre-processed for the filtering purpose. Later they are fed to SWT for the purpose of fusion using max/avg or any other fusion rule. Inverse SWT is applied on the on the fused coefficients, and thus final fused image is obtained.

By integrating the best aspects of SWT fusion rules with principal component analysis (PCA), a new medical image fusion method called a hybrid methodology is introduced in the article [3]. A more up-to-date image that provides the doctor with more information for a more accurate diagnosis is obtained by combining MRI and CT images. According to [4], a novel scheme for fusion is proposed, which is a blend of fourth-order partial differential equations (FPDE) and discrete cosine transform (DCT) in the SWT domain. The original input images should be first decomposed into their base and detail versions by using a technique based on the fourth-order partial differential equations. The final image is generated by taking the weighted average of the main components of the detail images. The basic images are fed to the SWT decomposition process.

The advantages and disadvantages of SWT method in case of image fusion are discussed in below Table 1.

Table 1. Advantage and Disadvantage of SWT method

Advantages	Disadvantages
Shift Invariance	Computational Complexity
Preserves Fine Details	Redundancy
Multi-Resolution Representation	Not Suitable for All Modalities
Noise Robustness	Possibility of Artifacts Introduction
Edge and Contrast Enhancement	Energy Loss in Reconstruction

B. NSCT Domain

Below Fig. 3 is the NSCT domain general image fusion modal. The NSCT processes two input images, A and B. The NSCT separates the input images into low-frequency (LFSB) and high-frequency sub bands (HFSB), respectively, for images A and B. The LFSB and HFSB are fused according to a predetermined fusion rule. To form a fused image, the combined LFSB and HFSB are subjected to the inverse NSCT. NSCT is applied on the source images as described in [5]. The fused LFSB coefficients are generated using the weighted average fusion rule, and the fused LFSB coefficients are obtained using an extractor that is based on the pre-trained VGG19 model. Lastly, the fused coefficients undergo an inverse NSCT to reconstitute the fusion results. To acquire the basic and detail layers, authors of [6] optimized for Tikhonov regularization using source images. Next, initial coefficient maps were created by sparsely decomposing detail layers using pre-trained dictionary filters after convolutional sparse representation processing. The fused detail layer was created by sparse reconstruction, and the fusion coefficient maps were refined using the NSCT domain's sub-band correlation. The averaging process was also employed to merge the base layers. The two-scale reconstruction was employed to create the final fused image.

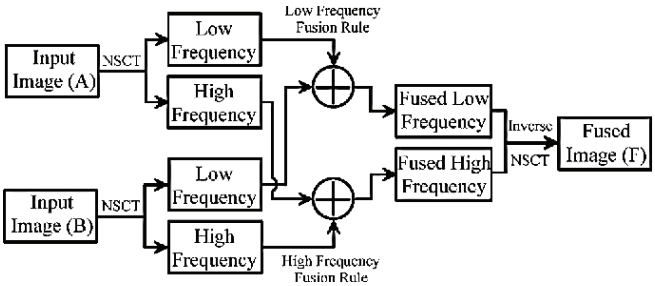


Fig 3. General image fusion modal using NSCT domain

The advantages and disadvantages of NSCT method in case of image fusion are discussed in Table 2.

Table 2. Advantage and Disadvantage of NSCT method

Advantages	Disadvantages
Multi-resolution and Multi-directional Analysis	Computational Complexity
	Consumes More Memory

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Shift-Invariance	Parameter Tuning
Edge Preservation	Difficulty in Interpretation
Sparse Representation	Possibility of Over-Smoothing
Reduced Artifacts	Lack of Standardization
Improved Visualization	

C. NSST Domain

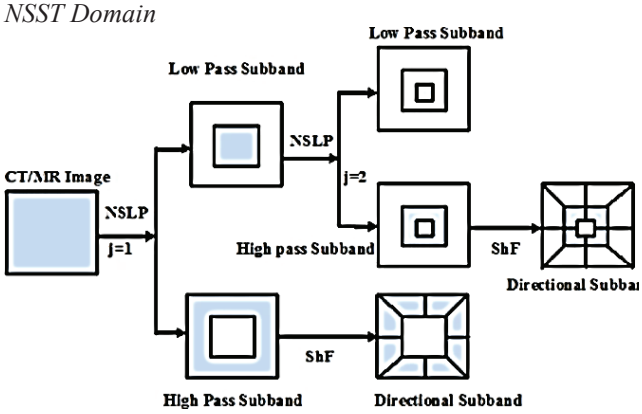


Fig 4. Decomposition of NSST on two levels

The breakdown of the NSST into LFSB and HFSB is shown in above Fig 4. The LPSB collects the image's large-scale, smooth information, while the HPSB catches finer details like edges and texture. Using NSST and SWT together, tha authors of [7] provide a new hybrid fusion strategy. The LFSB coefficients of NSST are fused using the regional energy as an activity parameter. The SWT is used to fuse the approximation sub-band of the NSST. As activity parameters, the maximum entropy of the squared coefficients and regional energy are used to fuse the LF and HF sub-bands of SWT, respectively. The purpose of this phase is to ensure that the image's brightness, contrast, edges, and texture are preserved. The integrated image produced by [8] includes data on both soft and hard tissues. Doctors can make more accurate diagnoses using the merged image. Use of the anisotropic diffusion technique and the NSST approaches are part of the suggested approach [8].

The advantages and disadvantages of NSST method in case of image fusion are discussed in below Table 3.

Table 3. Advantage and Disadvantage of NSST method

Advantages	Disadvantages
Multi-scale and Multi-directional Analysis	High Complexity and Computation Time
Shift-Invariance	Parameter Selection Dependent
Preservation of High-Frequency Details	Higher Memory Usage
Robustness to Noise	Artifact Introduction
Improved Contrast and Clarity	Limited Interpretability
Efficient in Multi-modal Fusion	Dependency on Fusion Rules

D. CNN Domain

To successfully fuse multimodal medical images, the research [9] proposed a deep learning CNN approach. One method that has been suggested uses a Siamese convolutional network to create a weight map that considers pixel mobility in various medical images. By using a multiscale method and medical image pyramids, the fusion process may achieve an accuracy level comparable to human visual recognition. A local comparison-based strategy is used to automatically update the decomposed coefficients. A novel rule for low-frequency adaptive synthesis was presented by the authors of [10]. An algorithm for synthesizing LFSB using the VGG19 network was suggested by them. To make medical image fusion more efficient, a new approach is developed. The general process of image fusion using CNN is depicted in Fig 5.

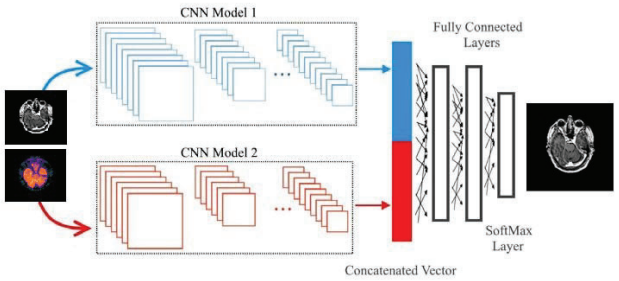


Fig 5. General CNN based image fusion process

The advantages and disadvantages of CNN method in case of image fusion are discussed in below Table 4.

Table 4. Advantage and Disadvantage of CNN method

Advantages	Disadvantages
Automated Feature Extraction	Data Dependency
Multimodal Data Integration	Overfitting Risk
Spatial Invariance	High Computational Cost
High Accuracy	Interpretability Issues
End-to-End Learning	Complexity in Network Design
Handles Complex Non-linear Relationships	Risk of Losing Detail
	Bias and Generalization

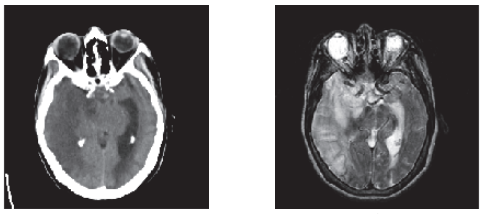
III. Results And Discussion

Figures 6 and 7 display datasets comprising CT, PET, and MRI images [10, 11]. Figures 8 and 9 present the visual quality results of various image fusion methods applied to these datasets. Notably, the edges and corners are well-preserved, particularly in the CNN-based method, which becomes more evident when the fused image is zoomed in. The texture remains uniform across the entire image, with no loss of information.

Since there is no reference image available for comparison, objective performance assessment metrics without a reference image are utilized to evaluate the fusion results across the different methods. The metrics used are Functional Mutual Information (FMI), Fusion Symmetry

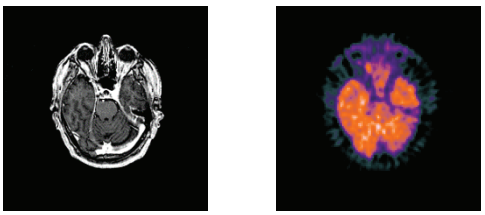
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(FS), Mean (M), Fusion Factor (FF), Standard Deviation (SD), Entropy (E), Fused Index (FI), and Sum of Correlation Difference (SCD). Except FS, all the metrics should be having higher values for the better results. Only FS should have lower value for the better results.



(a) (b)

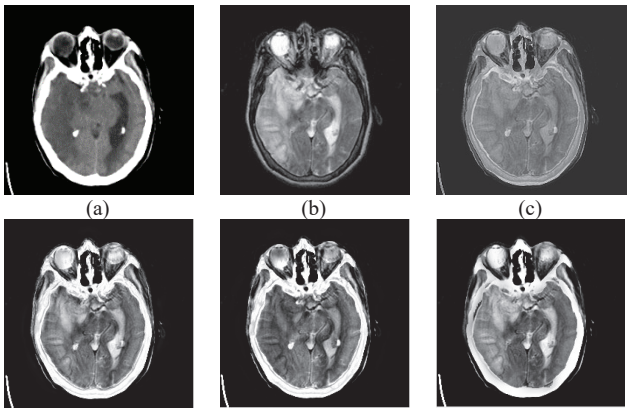
Fig 6. Dataset (a) CT image (b) MRI image



(a) (b)

Fig 7. Dataset(a) MRI image (b) PET image

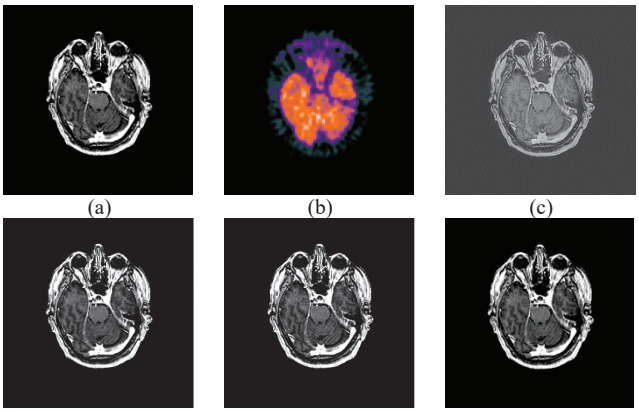
The visual results of some conventional methods as shown in Fig 8 and 9. The compared conventional methods are SWT, NSST, NSCT, and CNN.



(a) (b) (c)

(d) (e) (f)

Fig 8. (a) CT image (b) MRI image (c) SWT (d) NSCT (e) NSST (f) CNN



(a) (b) (c)

(d) (e) (f)

Fig 9. (a) MRI image (b) PET image (c) SWT (d) NSCT (e) NSST (f) CNN

Upon meticulous examination of the visual outcomes (Fig 8 and 9) and parametric data (Table 5, and 6), it can be concluded that the CNN based fusion method demonstrates superior performance relative to other compared methods evaluated. The comparative performance of CNN based fusion method surpasses mostly all the compared approaches.

Table 5. Average quantitative analysis of SCD, FS, FMI, and FF

Method	SCD	FS	FMI	FF
SWT	1.39	0.98	0.91	1.35
NSCT	1.45	0.99	0.98	1.71
NSST	1.43	0.91	0.95	1.72
CNN	**1.60**	**0.95**	**1.05**	**1.75**

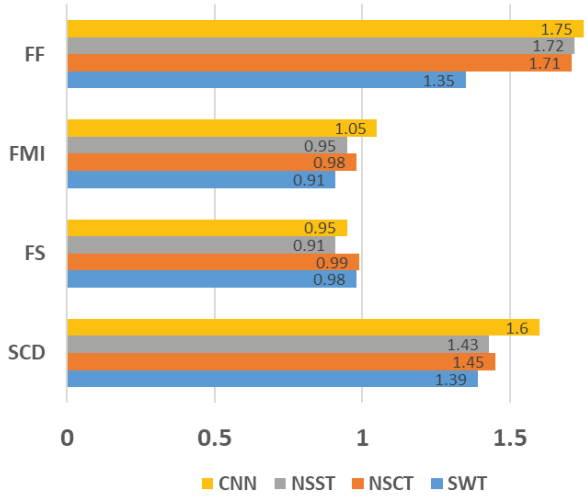


Fig 10. Graphical representation of Table 5

Table 6. Average quantitative analysis of SD, M, E, and FI

Method	SD	M	E	FI
SWT	40.90	19.52	1.38	20.60
NSCT	62.32	28.11	3.01	30.81
NSST	62.76	28.13	2.77	30.88
CNN	**63.12**	**29.05**	**3.10**	**31.21**

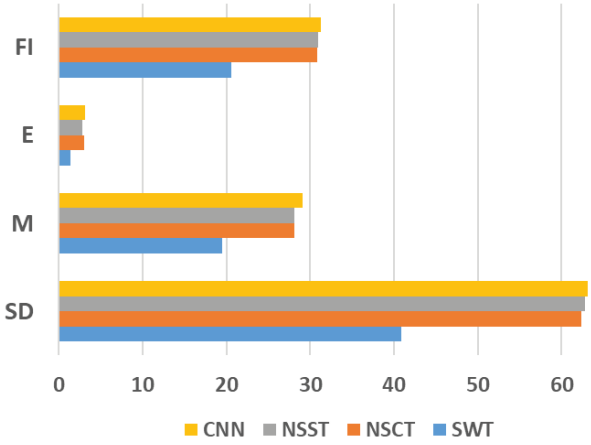


Fig 11. Graphical representation of Table 5

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The results presented in Tables 5 and 6 represent the average outcomes obtained from experiments conducted on more than 100 image datasets. These tables are graphically represented in Fig 10, and 11.

IV. Conclusion

The dataset for this study contains CT scan images, PET scan images, and MRI scans. This research investigates and compares many multimodal medical image fusion algorithms using SWT, NSST, NSCT, and CNN. The major emphasis is on both traditional and unconventional research in these areas, which are thoroughly explored. Furthermore, the research compares and analyses these methodologies utilizing several quantitative tests and image quality criteria. Overall, the visual and parametric quality of the fusion results generated by the CNN-based technique outperforms other methods, placing it first in the comparison. Because of its improved performance, the CNN-based technique is perfect for clinical decision-making, allowing clinicians to analyze diagnostic data with better precision using the fused images it creates.

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